Mean

1 Describe what the mean measures for a set of scores.

2 Find the mean of the following data sets:

a $8, 15, 6, 27, 3$

b $56, 89, 95, 71, 75, 84, 65, 83$

c $22.4, 25.4, 19.1, 24.3, 7.4$

d $-14, 0, -2, -18, -8, 0, -15, -1$

3 The mean of four scores is 21. If three of the scores are $17, 3$ and 8, find the fourth score.

4 The five numbers $16, 16, 17, 24, 17$ have a mean of 18. If a new number is added that is bigger than 24, will the mean be greater or smaller?

5 The five numbers $11, 13, 9, 13, 9$ have a mean of 11. If a new number is added that is smaller than 9, will the mean be greater or smaller?

6 The mean of a set of 41 scores is 18.6. If a score of 71.8 is added to the set, find the new mean. Round your answer to two decimal places.

7 Find the sum of the following sets of scores:

a A set of 29 scores with a mean of 37.7

b A set of 10 scores with a mean of 4

8 Calculate the number of scores in the following sets:

a The mean of a set of scores is 35 and the sum of the scores is 560.

b The mean of a set of scores is 38.6 and the sum of the scores is 694.8.

9 A statistician organized a set of data into the following frequency table:

a Complete the frequency distribution table.

b Calculate the mean, correct to two decimal places.

Score (x)	Frequency (f)	fx
5	14	
7	4	
9	2	
11	18	
13	6	
Totals		

10 For the following data set:

$$39,\ 42,\ 40,\ 39,\ 39,\ 40,\ 42,\ 38,\ 41,\ 42,\ 40,$$

$$39,\ 39,\ 42,\ 40,\ 39,\ 39,\ 38,\ 39,\ 38,\ 39,\ 40,\ 40$$

a Complete the frequency distribution table.

b Construct a column graph of the data.

c Calculate the mean, correct to one decimal place.

Score (x)	Frequency (f)	fx
38		
39		
40		
41		
42		
Total		

Median

11 Find the median of the following set of scores: $6, 8, 9, 11, 16, 17, 18$

12 State the position of the median in an ordered set of:

a 69 scores

b 152 scores

13 How many scores, n, are there in a data set if the number of scores is even and the median lies between the 24th and 25th scores.

14 For each of the following sets of scores:

i Sort the scores in ascending order.

ii Find the median.

a $69.4,\ 66.7,\ 46.6,\ 76.6,\ 52.8,\ 80.6,\ 63.9$

b $44.9,\ 45.6,\ -54.8,\ 74.7,\ -77.6,\ -42.6,\ 67.9,\ 40.6$

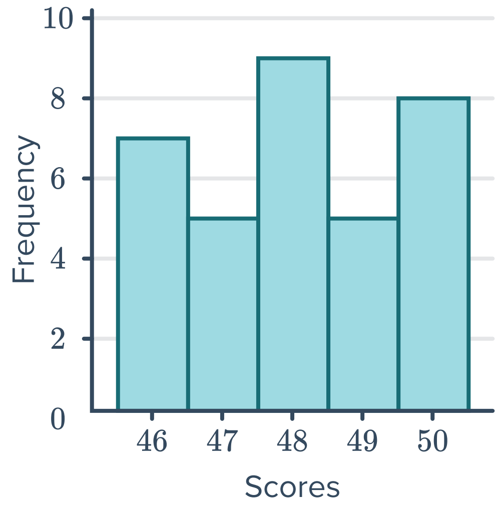
c $24,\ 11,\ 1,\ 3,\ 24,\ 28,\ 16,\ 16$

15 Find 4 consecutive odd numbers whose median is 40.

16 For the given histogram:

a Find the number of scores.

b Find the median.



17 For each of the following column graphs, find:

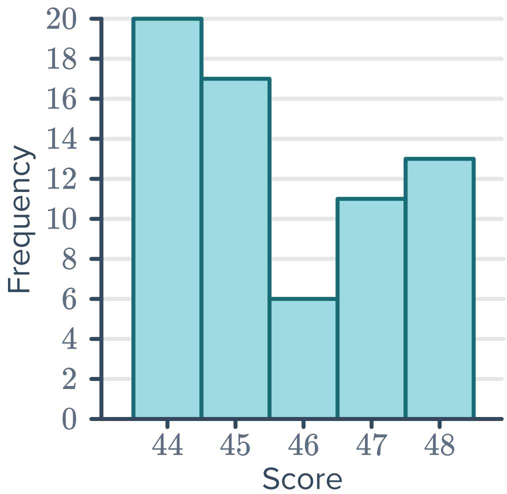
i The total number of scores.

ii The median.

iii The total sum of the scores.

iv The mean, correct to one decimal place.

a



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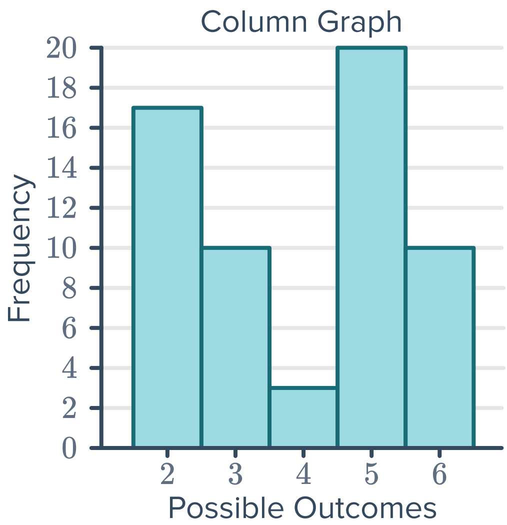


18 Find the median from the following stem and leaf plot:

	Leaf
6	2 3
7	2 4 9
8	3 4 9 9
9	1 1 5

Key: $8|3 = 83$

19 For each frequency distribution table, find [redacted]edian of the scores.

a

Score	Frequency
23	2
24	26
25	37
26	24
27	25

b

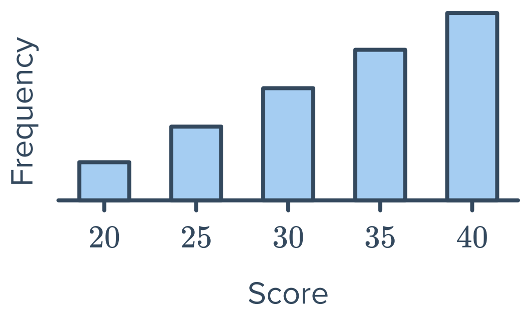
Score (x)	Frequency (f)
10	6
14	9
18	6
21	7
24	8

c

Score	11	12	13	14	15
Frequency	24	9	21	9	6

20 Determine whether the following sets of data have equal median and mean:

a



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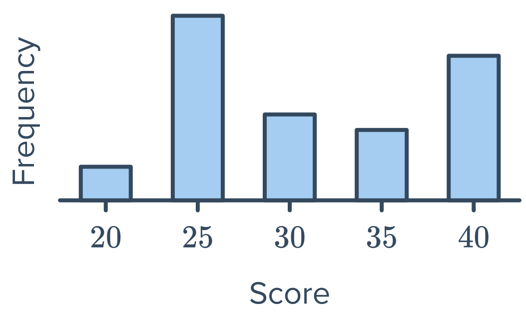
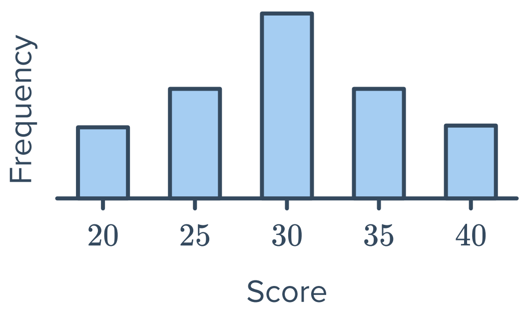
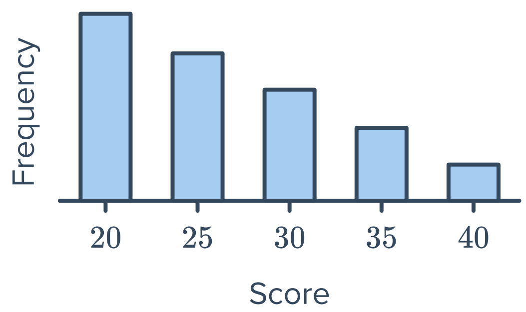


c



d



Mode

21 Find the mode(s) of the following data sets:

a $2, 2, 6, 8, 8, 8, 8, 8, 12, 14, 14, 14, 14, 14, 18, 18$

b

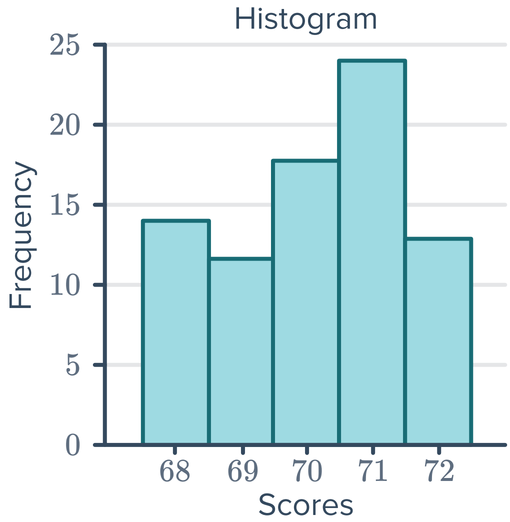
Score	Frequency
25	17
26	42
27	35
28	32
29	12
30	20

c

	Leaf
2	2 5 6 7 9
3	0 0 5 6 8
4	0 1 3 8 9

Key: $2|4 = 2.4$

d



22 The following stem plot shows the number of hours students spent studying for a science exam.
Find the following, correct to two decimal places if necessary:

a Mean

b Median

c Mode

	Leaf
6	0 2 3 6 7 7 8
7	3 6 9
8	0 5 6 7
9	2

Key: $1|2 \ = \ 12$

Measures of center

23 Determine the measure of center that should be used for the following data sets. Explain your choice.

a $12, \quad 15, \quad 16, \quad 21, \quad 22, \quad 25$

b $85, \quad 85, \quad 95, \quad 70, \quad 60, \quad 70, \quad 105$

c $36, \quad 46, \quad 49, \quad 40, \quad 52, \quad 44, \quad 39, \quad 37$

d $6, \quad 12, \quad 18, \quad 9, \quad 0, \quad 3, \quad 15, \quad 46$

24 Find the most appropriate measure of center for the following scenarios:

a For a particular street, house prices for houses sold in the last 36 months (in 1000's) were given:

$$920, \quad 920, \quad 709, \quad 694, \quad 667, \quad 657, \quad 630, \quad 610, \quad 584, \quad 567, \quad 555$$

b The number of animal races that were won by a trainer over the years shown are listed in the table:

Year	2003	2004	2005	2006	2007
Races won	116	105	102	108	113

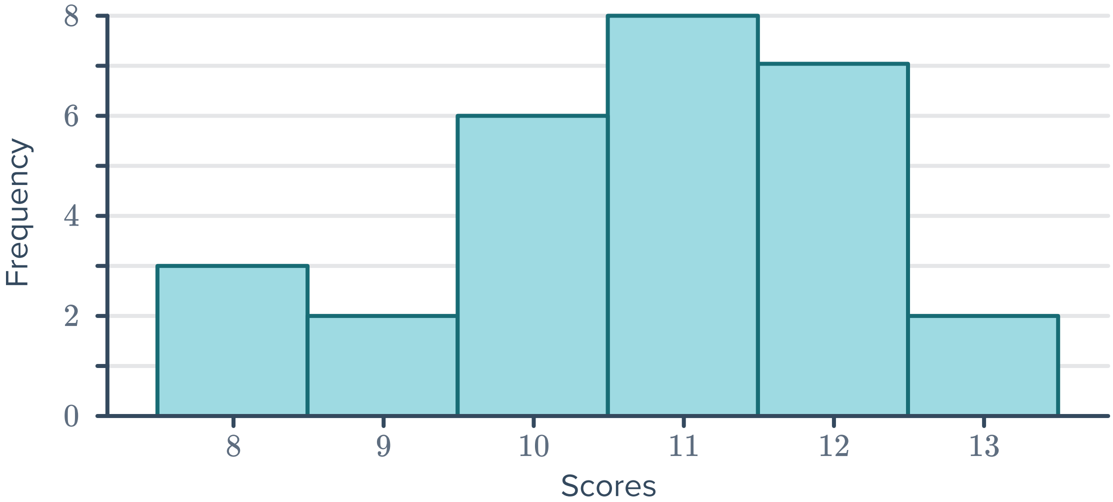
25 A consumer group surveys the price of gas per gallon at six different gas stations and the results are as follows:

$$\$1.68, \$1.64, \$1.64, \$1.71, \$1.64, \$1.57$$

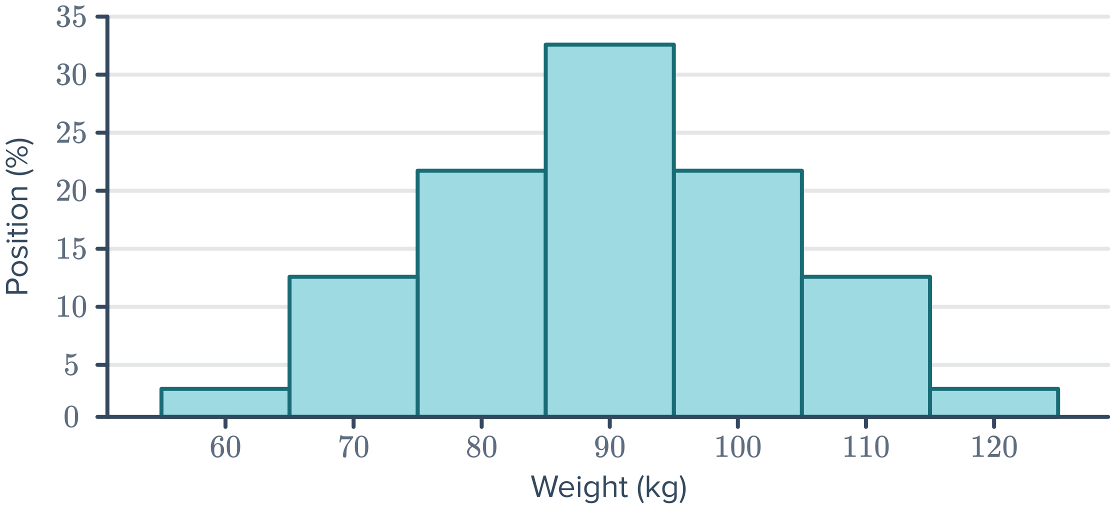
State the mode gas price.

26 Find the most appropriate measure of cen[illegible]r the following data set:

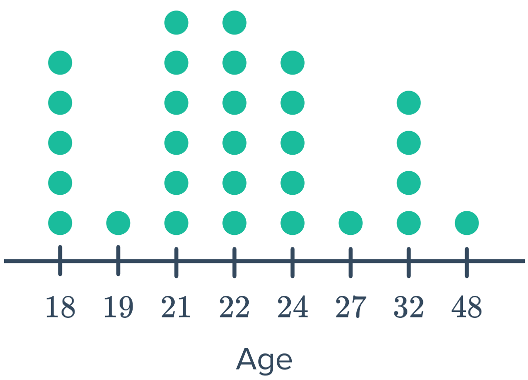
a



b



c



Outliers

27 If a data set has a mode of x and an outlier, what will happen to the mode if the outlier is removed?

28 A set of data has a mean of x. When the outlier is removed the mean rises. What does this imply for the value of the outlier?

29 The selling price of recently sold houses are:

$$\$467\,000,\ \$413\,000,\ \$410\,000,\ \$456\,000,\ \$487\,000,\ \$929\,000$$

a   Find the mean selling price, to the nearest thousand dollars.

b   Which of the selling prices raises the mean so that it is not reflective of most of the prices?

c   Recalculate the mean selling price excluding this outlier.

Applications

30 The frequency table below shows the resting heart rate of some people taking part in a study:

Heart Rate	Class center (x)	Frequency (f)	$f \times x$
$30 - 39$		13	
$40 - 49$		22	
$50 - 59$		24	
$60 - 69$		36	

a   Complete the table.

b   What is the mean resting heart rate? Round your answer to two decimal places.

31 The table shows the scores of Student A and Student B in five separate tests:

a Find the mean score for Student A.

b Find the mean score for Student B.

c What is the combined mean of the scores of the two students.

d What is the greatest score overall? Which student obtained that score?

e What is the least score overall? Which student obtained that score?

Test	Student A	Student B
1	97	78
2	87	96
3	94	92
4	73	72
5	79	86

32 A real estate agent wanted to determine a typical house price in a certain area. He gathered the selling price of some houses (in dollars):

$$317\,000,\ 320\,000,\ 347\,000,\ 360\,000,\ 378\,000,\ 395\,000,\ 438\,000,\ 461\,000,\ 479\,000,\ 499\,000$$

a Find the mean house price.

b What percentage of the house prices exceeded the mean?

c Find the median house price.

d What percentage of house prices exceeded the median?

33 Han wants to try out as a batsman for a cricket team. In his last three matches, he scored $61, 75$ and 66 runs. In his last match before trying out, he wants to lift his mean to 70. If x is the number of runs he needs to score to achieve this, find the value of x.

34 A teacher calculated the mean of 25 students' scores to be 64. A student who later completed the assessment got a score of 55. Find the new mean of the class, correct to two decimal places.

35 Carl has been recording his spelling test scores for the past semester:

$$14,\ 16,\ 2,\ 15,\ 15,\ 16,\ 15$$

a Calculate the median of Carl's scores.

b Calculate the mean of Carl's scores. Round your answer to two decimal places.

c Which measure of center most accurately describes the center of this data set? Explain your answer.

36 The following stem plot shows the ages o[redacted]mployees in a company:

a How many of the employees are in their 30's?

b What is the age of the oldest employee?

c What is the age of the youngest employee?

d Find the median age of the employees.

e What is the modal age group?

	Leaf
2	0 2 3 4 6 6 7
3	1 5 6 7 8
4	0 3 4 6 7
5	0 0 8

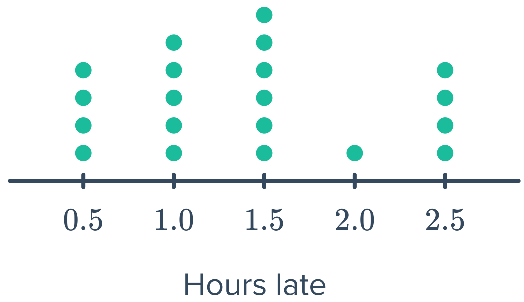
Key: $1|2 \;=\; 12$ years old

37 On Sunday, 20 planes were delayed at the airport. The dot plot shows the number of hours each departure was delayed:

a What was the median number of minutes a plane was delayed?

b What percentage of planes were delayed for longer than the median time?

c If a plane is delayed for more than 30 minutes, the airline must pay $\$5000$. In total, how much were airlines fined that day?



38 The luggage check-in weight (in kilograms) of passengers is given in the following frequency table:

a Complete the table.

b Find the median check-in weight.

c Find the mean check-in weight, correct to two decimal places.

d When one of the passengers sees that the weight of their luggage is 21 kilograms, they decide to add some items to the bag. This changes the mean or the median?

Weight (x)	Frequency (f)	fx
16	12	
17	17	
18	21	
19	10	
20	18	
21	22	
Total		

39 Durations of calls (in minutes) made in a h[redacted]old were recorded as follows:

$$5,\ 11,\ 3,\ 9,\ 5,\ 14,\ 5,\ 14,\ 14,\ 3,\ 7,\ 7,\ 7,\ 5,\ 3,\ 3,\ 7,\ 14,\ 5$$

a Find the total number of calls made.

b Find the longest duration of a call.

c Find the shortest duration of a call.

d Find the mean duration of a call. Round your answer to two decimal places.

e Find the modal duration.

f Find the median duration.